Notes 08/26
Precedence of logical operations

()	0
$\neg$	1
$\wedge$	2
$\vee$	3
$\to$	4
$\leftrightarrow$	5

Example.

p	q	$\neg p$	$\neg q$	$p \vee \neg q$	$\neg p \wedge \neg q$	$(p \vee \neg q) \to (\neg p \wedge \neg q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	F	F	T
F	F	T	T	T	T	T

Example.

p	q	r	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \to r$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	T	F	T
F	F	F	T	F	T

If a proposition's premise is false, it is a vacuously true proposition.

Logic puzzles

Angel tells the truth. Demon always lies.

"Belial is an angel only when Melodi isn't an angel."

b = Belial is an angel. m = Melodi is an angel.

b	m	$\neg m$	$b \to \neg m$	$b \leftrightarrow (b \to \neg m)$	$m \wedge \neg b$
T	T	F	F	F	F
T	F	T	T	T	F
F	T	F	T	F	T
F	F	T	T	F	F

$m \leftrightarrow (m \wedge \neg b)$

3 chests. 1 treasure, 1 truthful chest.

Chest 1: "empty chest." Chest 2: "empty chest." Chest 3: "in chest 2."

p = treasure is in chest 1 (chest 1's claim is $\neg p$)
q = treasure is in chest 2 (chest 2's claim is $\neg q$)
r = treasure is in chest 3 (chest 3's claim is q)

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Chest 2 is truthful; chest 1 has the treasure.